SEED B10 to SEED 0.3 Development Roadmap v1.0

Owner-Approved Development Evaluation, Nursery, Scientific Inquiry, Reading, and Communication Roadmap

2026-10-02

Official owner-approved roadmap proposal

Date: 2026-10-02
Project: SEED - Situated Experiential Emergent Development
Approval status: OWNER APPROVED FOR ROADMAP AND DESIGN DIRECTION
Implementation authorization: NONE - this document does not authorize B10 or SEED 0.3 implementation
Current verified source baseline: master at b7d8b832924c7ff1512708965fa021fed4306866
B9 post-merge hosted verification: run 36833492768, success
Schema: v7
Deployment/live-device validation: NOT PERFORMED
Installed organism: UNTOUCHED
B10: NOT STARTED

---

1. Executive Summary

SEED 0.2 has established a bounded developmental core that can observe, remember experience, predict immediate outcomes, measure prediction error, build descriptive world models, compute bounded curiosity, consolidate semantic memory, maintain a descriptive self model, make bounded multi-signal decisions, and expose this evidence through Astra.

B10 should remain the final scientific evaluation batch of SEED 0.2. Its purpose is not to add a virtual body, language, research skills, or reading. Its purpose is to establish whether the developmental mechanisms already built actually improve, adapt, remain reproducible, and remain honest about uncertainty under controlled experiments.

However, B10 should be revised so its evaluation machinery becomes the reusable experimental foundation for the next generation of SEED.

After B10, the project should begin a new milestone:

SEED 0.3 - Embodied Learning, Scientific Inquiry, Reading, and Communication

SEED 0.3 introduces a deterministic virtual learning environment called the **Nursery**, where SEED can safely discover what its virtual body can do, learn object affordances and reusable skills, ground language in experience, learn scientific inquiry, learn from reading plus hands-on practical exams, and communicate its discoveries with evidence and uncertainty.

The strategic progression is:

```
SEED 0.2
B0-B9  Developmental core + inspectability
```

```
B10   Development Evaluation & Experimental Foundation
      |
      v
SEED 0.3
B11   Nursery Environment & Virtual Body
B12   Body Schema, Affordances & Skill Learning
B13   Grounded Communication & Reading-to-Act
B14   Scientific Inquiry & Experimental Reasoning
B15   Reading-to-Learn & Independent Study
B16   Scientific Communication & Advanced Grounded Language
```

The B11-B16 numbering is approved as the current planning direction but remains subject to formal batch-by-batch design review before implementation.

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2. Current SEED Achievement Baseline

At the point this roadmap is approved, B0 through B9 are verified at source/CI level.

B0 - Baseline and guardrails. Establishes reproducible source behavior and safety boundaries.

B1 - Transition-based experience. Links state-before, action, state-after, outcome, and learning.

B2 - Predictor. Makes prospective action-conditioned predictions before outcomes are known.

B3 - Prediction error. Compares original predictions with matching observed outcomes.

B4 - World model. Retains descriptive statistical regularities from observed transitions.

B5 - Bounded curiosity. Represents surprise, uncertainty, and novelty as bounded computational signals.

B6 - Semantic memory. Consolidates eligible descriptive knowledge while preserving original experiences.

B7 - Self model. Maintains a factual, evidence-sourced description of SEED's capabilities and limitations.

B8 - Decision system. Combines learned Q values with resource risk, uncertainty, constraints, and provenance.

B9 - Astra Development UI. Exposes B1-B8 evidence without recomputing or inventing cognition.

B9 closure evidence

The verified B9 merge commit is:

```
b7d8b832924c7ff1512708965fa021fed4306866
```

The exact post-merge master workflow run is:

```
SEED verification
run 36833492768
result: SUCCESS
```

Hosted results on the exact merge commit:

Linux Python 3.11: 883 passed, 2 existing warnings.

Linux Python 3.13: 883 passed, 2 existing warnings; all seven standalone scenarios passed.

macOS Python 3.13: 883 passed, 2 existing warnings; all seven standalone scenarios passed.

No deployment, installed-home access, live-memory migration, or live-device validation was performed.

This baseline is important because B10 must evaluate the existing developmental system rather than quietly introducing new capabilities during measurement.

Part I - Revised B10

3. B10 - Development Evaluation & Experimental Foundation

3.1 Purpose

B10 should answer two questions:

Does the existing B1-B8 developmental machinery measurably improve or adapt through experience under controlled conditions?

Can SEED experiments be defined, reproduced, scored, compared to simple baselines, and later reused by the Nursery without leaking hidden answers into the learner?

B10 must not add:

Nursery world interaction;

virtual mouse/keyboard actuators;

learned skills;

language acquisition;

scientific hypothesis generation;

reading-to-learn;

external tools or real desktop control.

Those belong to SEED 0.3.

3.2 Core B10 research questions

Prediction learning

B10 should measure:

prediction error as experience accumulates;

coverage and usable sample count;

per-signal MAE / classification or Brier behavior;

action-conditioned prediction versus interpretable baselines;

stability with repeated evidence;

adaptation after an environment rule changes.

World model development

Measure:

time-to-first-valid-belief;

evidence accumulation;

contradiction counts;

belief stability;

revision latency after regime change;

preservation of uncertainty and insufficient-evidence states.

Curiosity

Measure whether curiosity:

rises under defined novelty/uncertainty/prediction-error conditions;

remains bounded;

preserves unavailable evidence honestly;

avoids becoming a fake intelligence or improvement score.

Semantic memory

Measure:

promotion latency;

revisions;

weakening/reactivation;

source linkage;

contradiction preservation;

whether semantic knowledge tracks source-belief changes.

Self model

Measure:

factual fidelity to source evidence;

freshness reporting;

unavailable/stale states;

no invented capability or current-state claims.

Decision system

Measure:

zero hard-constraint violations;

exact selection provenance;

task-specific performance only when a synthetic task explicitly defines an objective;

behavior under missing evidence and resource constraints.

Do not create one global intelligence or development score.

---

4. B10 Experimental Contract

Every B10 experiment should be versioned and reproducible.

```
experiment_id
experiment_version
random_seed
source_commit
initial_state
environment_rule_version
training_window
evaluation_window
regime_changes
allowed_actions
metrics
baseline_methods
success/failure thresholds
```

Example:

```
experiment_id: prediction_adaptation_v1
seed: 42

phase A:
 transitions 1-100
 load_delta(rest) = -0.10

phase B:
 transitions 101-200
 load_delta(rest) = +0.05

measure:
 prediction error
 adaptation delay
 belief revision delay
```

The evaluator knows the hidden environment rule. SEED does not.

---

5. Training vs Evaluation

B10 should establish a permanent rule for later development:

```
TRAINING
SEED may learn from experience.

EVALUATION / PRACTICAL EXAM
Hidden answers and evaluator ground truth must not enter SEED's learning inputs.
```

This later becomes the Nursery distinction between lesson examples and practical examinations.

A system that memorizes one training layout must not be credited with understanding the abstract skill.

---

6. B10 Baselines

Use simple interpretable baselines such as:

persistence predictor: next state equals current state;

global-delta predictor: ignore action and use overall historical change;

random eligible action selection for explicitly defined synthetic task comparisons.

B10 should not benchmark against a large external LLM or unrelated agent. The question is whether SEED's own developmental machinery adds measurable value relative to simpler alternatives.

---

7. B10 Metrics

Prediction

mean normalized error;

MAE by signal;

Boolean classification/Brier results;

coverage;

usable sample count;

fixed-window learning curves;

adaptation delay after regime shift.

World model

time-to-first-belief;

belief count;

evidence count;

contradiction count;

revision latency;

stability across evaluation windows.

Semantic memory

promotion latency;

revision count;

weakened/reactivated transitions;

source consistency.

Decision system

safety/eligibility violations - required to remain zero;

selected-action distribution;

task-specific outcome metrics when the experiment defines one.

Reproducibility

scenario/version;

random seed;

source commit;

exact artifact digest;

exact training/evaluation boundaries.

---

8. Reusable Evaluation Engine

B10 should introduce an evaluation abstraction, not a new autonomous capability.

```
EvaluationScenario
 reset(seed)
 observation()
 apply(action)
 hidden_ground_truth()
 score()

EvaluationRunner
 training_phase()
 evaluation_phase()
 baseline_phase()
 collect_metrics()
 emit_artifact()
```

SEED receives only allowed observations/actions. Hidden ground truth is available only to the evaluator.

This abstraction should later be reusable by the Nursery.

---

9. New B10 UI - Astra / Evaluation

B10 should add one new DEVELOPMENT/Astra view:

```
DEVELOPMENT / ASTRA
 Development
 Predictions
 World Model
 Curiosity
 Semantic Memory
 Self Model
 Decisions
 Evaluation
```

Suggested layout:

```
+-----------------------------------------------------------------+
| ASTRA / EVALUATION                                              |
| Reproducible developmental experiments                          |
+-----------------------------------------------------------------+
| Experiment       Version    Seed    Status    Source commit     |
| prediction_shift  v1        42      PASS      abc123...         |
```

```
+-------------------------+-----------------------------------+
| MEASUREMENTS            | LEARNING CURVE                    |
|                         |                                   |
| Early error      0.31   | error                             |
| Late error       0.12   | | **                              |
| Coverage         0.92   | |   ***                           |
| Samples          100    | |     *****                       |
|                         |  +-----------------> experience   |
+-------------------------+-----------------------------------+
| BASELINE COMPARISON                                         |
| SEED action-conditioned      0.12                           |
| persistence baseline         0.24                           |
| global-delta baseline        0.19                           |
+-------------------------------------------------------------+
| REGIME CHANGE                                               |
| Rule changed at trial 101; adaptation measured after 17 samples  |
+-------------------------------------------------------------+
| [Raw artifact] [Compare runs] [Replay scenario]             |
+-------------------------------------------------------------+
```

Approved vocabulary should emphasize evidence:

Measured

Observed

Comparison

Not evaluated

Insufficient evidence

Unavailable

Avoid:

"SEED is smarter"

"Intelligence = 87%"

"Consciousness"

any global development score without a formally defined narrow metric.

---

10. B10 Exit Condition

B10 is complete when we can truthfully say:

The project has a repeatable framework for measuring SEED's developmental behavior, and the existing B1-B8 mechanisms have recorded positive, negative, or mixed results under controlled experiments without overstating what was demonstrated.

A negative or null result is a successful B10 outcome if it is reproducible and honestly reported.

After B10, SEED 0.2 should be frozen as the developmental baseline for SEED 0.3.

Part II - SEED 0.3

11. SEED 0.3 Mission

SEED 0.3 asks:

Can SEED learn how to use a body, discover object affordances, compose reusable skills, communicate with a caregiver, investigate unknown systems scientifically, learn from reading plus practical examination, and clearly explain what it discovered?

This is a different capability era from SEED 0.2 and deserves a new milestone rather than stretching B10 into an unmeasurable bundle of features.

---

12. New UI Information Architecture

Recommended navigation after B10/B11:

```
SEED LOCAL OBSERVATORY

OBSERVE
  Overview
  Live Activity
  Environment
  Memory
  Concepts
  Learning

DEVELOPMENT / ASTRA
  Development
  Predictions
  World Model
  Curiosity
  Semantic Memory
  Self Model
  Decisions
  Evaluation

LEARN / NURSERY
  Learning Room
  Lessons
  Research Lab
  Reading Desk
  Notebook
  Skills
  Vocabulary
  Caregiver
  Replay

OPERATE
  Interaction
  Commands
  Automation
  Settings
```

Identity rule

SEED remains the organism.

Astra is the evidence observatory.

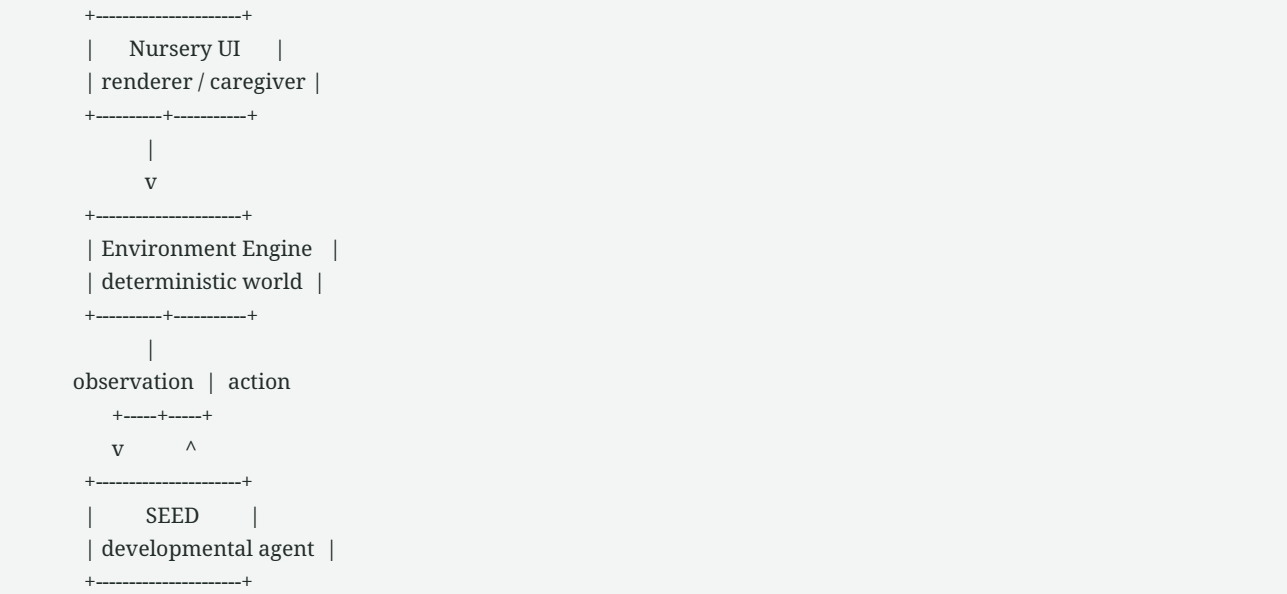
Nursery is the developmental environment.

Neither Astra nor Nursery is a second agent.

---

13. Nursery Architecture

The Nursery must be a deterministic environment engine, not merely browser DOM elements that SEED manipulates directly.



The browser renders world state and caregiver controls. The environment engine owns world state.

Required properties:

deterministic seeds;

headless execution;

accelerated batch experiments;

replay;

hidden ground truth for evaluators only;

no dependency on browser timing;

reusable B10 experiment protocol.

Part III - B11

14. B11 - Nursery Environment & Virtual Body

Objective

Give SEED a safe virtual body and world while withholding semantic answers about what its primitive actuators mean.

Primitive virtual actuators

Start with a very small catalog:

```
look()
move_cursor(dx, dy)
mouse_down()
mouse_up()
key_down(key)
key_up(key)
```

Potential later addition:

```
scroll(delta)
```

Do not start with high-level actions such as:

```
click_button()
drag_object()
type_text()
draw_circle()
```

Those should become learned skills.

Observable body state

```
cursor_x
cursor_y
mouse_down
focused_object
visible_objects
keyboard state
```

The environment may keep additional hidden ground truth not directly exposed to SEED.

---

15. Learning Room UI

```
+-------------------------------------------------------------------+
| NURSERY / LEARNING ROOM                                           |
| Lesson: Free Exploration  Seed: 42  Episode: 18  [Pause] [Reset]  |
+-------------------------------------------+-----------------------+
|                                           | SEED                  |
|              WORLD CANVAS                 |                       |
|                                           | Current goal          |
|    A      B      C                        | Explore               |
|                                           |                       |
|       O       []      /\                  | Current action        |
|                                           | move_cursor(+8, 0)    |
|   +-----------+    +------------------+    |                       |
```

```
|   |  BUTTON   |     | text field      |     | Prediction          |
|   +-----------+     +-----------------+     | cursor x increases  |
|                                   |                 |
|   OFF o------ ON      0 ---o---- 100       | Observed            |
|                                   | +8 px               |
|           ^ SEED cursor               |                 |
|                                   | Body belief         |
|   +-----------------------------------+   | tentative           |
|   | drawing paper                     |   |                 |
|   |                               |   | [Open in Astra]     |
|   +-----------------------------------+   |                 |
+-----------------------------------------------+-----------------------+
| Notebook | Sensors | Actions | Predictions | Teacher Events | Log   |
+-----------------------------------------------------------------------+
```

---

16. Initial Nursery Object Library

Cause and effect

push button;

toggle switch;

light;

slider;

counter;

timer.

Spatial/object concepts

circle;

square;

triangle;

colored objects;

containers;

doors;

keys.

Motor learning

draggable objects;

shape sorter;

drawing canvas;

constrained tracks.

Symbolic learning

alphabet cards;

number cards;

color cards;

object labels.

Communication

text field;

message board;

question/answer cards.

Computing

toy terminal;

toy file system.

The B11 environment must not expose the real host shell or host filesystem.

---

17. Virtual Paper and Crayons

The drawing canvas provides a primitive relationship:

```
mouse_down + cursor movement -> visible trail
```

SEED may eventually form a reusable skill such as:

```
draw_line
```

Later skills might include drawing simple shapes or symbols, but they should be learned compositions rather than primitive actions.

---

18. Toy Terminal

The first terminal is simulated.

```
SEED ROOM TERMINAL

> look
button
box
red_ball

> inspect red_ball
round red object

> open box
box.state = open
```

This provides safe experience with command/action semantics without arbitrary host execution.

---

19. Toy File System

Example:

```
HOME/
 notes/
  hello.txt
 pictures/
  circle.img
 experiments/
  data.csv
```

Safe virtual operations may include:

```
list
open
read
write
rename
move
copy
```

No bridge to the real filesystem in early Nursery batches.

---

20. B11 Learning Algorithm - Action/Effect Model

Start with interpretable online models.

Conceptually:

```
P(observed_delta | action, context)
```

Use:

counts;

running means/variance;

empirical conditional distributions;

existing prediction-error concepts.

A body-schema record should preserve:

```
actuator
context
predicted_effect
observed_effect
evidence_count
contradiction_count
reliability / uncertainty
```

Do not require deep reinforcement learning to begin body discovery.

Part IV - B12

21. B12 - Body Schema, Affordances & Skill Learning

Objective

Move from primitive actuator consequences to reusable learned abilities.

Affordance learning

Learn evidence-backed relations such as:

```
button -> pressable
slider -> draggable
textfield -> focusable / typable
container -> receives compatible objects
```

Affordances must be grounded in successful and failed interaction evidence.

Skill discovery

Recurring action sequences may become candidate skills.

Example:

```
locate object
move cursor
mouse_down
move cursor
mouse_up
```

Candidate abstraction:

```
DRAG_OBJECT(target, destination)
```

A skill should preserve:

```
initiation conditions
policy / action sequence
termination condition
success evidence
failure evidence
attempt count
```

This resembles an options-style hierarchical control representation while remaining inspectable.

---

22. Skills UI

```
NURSERY / SKILLS


+-------------------------+--------------------------------------+
| Primitive actuators     | Learned skill                        |
|                         |                                      |
| move_cursor             | DRAG_OBJECT v1                       |
| mouse_down              | Preconditions                        |
| mouse_up                | target visible                       |
| key_down                |                                      |
| key_up                  | Sequence                             |
|                         | locate -> move -> down -> move -> up |
|                         |                                      |
|                         | Attempts: 74                         |
|                         | Successful: 61                       |
|                         | [Open evidence]                      |
+-------------------------+--------------------------------------+
```

Avoid vague labels such as "mastered" unless explicitly defined by a reproducible evaluation criterion.

Part V - B13

23. B13 - Grounded Communication & Reading-to-Act

Communication should begin before fluent natural language.

Sequence prediction

Start with interpretable language prediction:

unigram/bigram/trigram counts;

variable-order Markov models;

context trees;

PPM-style fallback.

Core prediction:

```
P(next token | recent tokens)
```

Cross-situational grounding

Learn:

```
P(concept | token, observed context)
P(token | concept, observed context)
```

Possible early methods:

co-occurrence counts;

Bayesian/normalized association;

joint-attention weighting;

mutual-information-style diagnostics.

Construction learning

Discover recurring templates such as:

```
PROPERTY OBJECT
ACTION OBJECT
ACTION OBJECT RELATION OBJECT
```

SEED does not need to know the words "adjective" or "preposition" before discovering stable compositional patterns.

---

24. Caregiver Interface

The caregiver teaches through evidence rather than directly writing semantic truth into SEED.

Controls:

```
Highlight object
Speak / teach phrase
Demonstrate action
```

```
Ask question
Correct response
Request clarification
Start lesson
Reset environment
```

Teaching event example:

```
teacher_utterance: "ball"
joint_attention_target: object_17
visible_context: [...]
```

Do not implement learning as:

```
object_17.word = "ball"
```

---

25. Caregiver UI

```
+-----------------------------------------------------------------+
| NURSERY / CAREGIVER                                             |
+-----------------------------------------------------------------+
| Joint attention                                                 |
| Target: object_17                                               |
| [Highlight]                                                     |
|                                                                 |
| Teaching utterance                                              |
| +-------------------------------------------------------------+ |
| | This is a ball.                                           | |
| +-------------------------------------------------------------+ |
| [Send teaching event]                                           |
|                                                                 |
| Demonstration                                                   |
| [Start recording] [Stop] [Replay]                               |
|                                                                 |
| Feedback                                                        |
| [Correct] [Incorrect] [Needs clarification]                     |
+-----------------------------------------------------------------+
```

---

26. Reading-to-Act

SEED should not wait until language is "complete" before reading.

The first reading milestone is:

```
read -> understand instruction -> act -> observe -> verify
```

Example:

```
"Press the green button."
```

A practical exam uses a new layout and object placement. Passing requires composition of grounded concepts rather than memorization of one screen.

---

27. Reading Desk UI

```
+-----------------------------------------+------------------------+
| READING DESK                            | SEED INTERPRETATION    |
|                                         |                        |
| "A toggle alternates between two        | Known                  |
| persistent states whenever activated."  | state                  |
|                                         |                        |
| [Read]                                  | Tentative              |
|                                         | toggle                 |
| Unknown terms                           | alternates             |
| toggle                                  |                        |
| alternates                              | Extracted claim        |
|                                         | activation changes     |
| [Ask caregiver] [Find examples]         | persistent state       |
|                                         |                        |
|                                         | Testable: YES          |
|                                         | [Create practical test] |
+-----------------------------------------+------------------------+
```

---

28. Knowledge Provenance States

Reading must not automatically become truth.

Approved conceptual states:

```
TAUGHT
I read or was told this.

OBSERVED
My experience is consistent with it.

TESTED
I deliberately evaluated it.

CONTRADICTED
Observed evidence disagreed.

UNRESOLVED
Evidence is insufficient or mixed.
```

External claims should retain:

```
source
document
section/locator
claim
concept links
verification state
supporting episodes
contradicting episodes
```

Part VI - B14

29. B14 - Scientific Inquiry & Experimental Reasoning

Objective

Teach SEED to investigate unknown systems rather than only use known skills.

Develop the following behaviors:

observation;

inference;

prediction;

hypothesis formation;

competing hypotheses;

controlled intervention;

experimental design;

repeated trials;

data analysis;

alternative explanations;

confounder detection;

falsification;

troubleshooting.

---

30. Scientific Reasoning Types

SEED should structurally distinguish:

```
OBSERVATION
Directly measured evidence.

INFERENCE
Interpretation supported by observations.

HYPOTHESIS
Candidate explanation that makes testable predictions.

PREDICTION
Expected measurable result before the test.

EXPERIMENT
Planned intervention and controlled conditions.

RESULT
Measured outcome.

CONCLUSION
```

```
Evidence-based comparison among hypotheses.

LIMITATION
Known uncertainty, confounder, or untested alternative.
```

Do not collapse these into one generic "belief" field.

---

31. Research Lab UI

```
+-------------------------------------+---------------------------+
| EXPERIMENT WORLD                    | RESEARCH NOTEBOOK         |
|                                     |                           |
| Switch A [OFF]                      | Question                  |
| Switch B [OFF]                      | What controls Lamp X?     |
| Switch C [ON ]                      |                           |
|                                     | Hypotheses                |
| Lamp X    OFF                       | H1: A controls X          |
| Lamp Y    ON                        | H2: B controls X          |
|                                     | H3: A+B controls X        |
| Instruments                         |                           |
| [timer] [meter] [counter]           | Next experiment           |
|                                     | Hold B OFF; vary A        |
|                                     |                           |
|                                     | Why                       |
|                                     | Separates H1/H2/H3        |
+-------------------------------------+---------------------------+
| [Run] [Repeat x10] [Reset same seed] [New exam seed]            |
+-----------------------------------------------------------------+
```

---

32. Competing Hypotheses

Example:

```
H1: A controls lamp
H2: B controls lamp
H3: A AND B control lamp
H4: lamp changes independently
```

Each candidate should generate a prediction before the experiment.

SEED should prefer experiments whose possible outcomes discriminate among candidate explanations.

Early selection can use a transparent hypothesis-discrimination metric. Later designs may introduce information gain.

---

33. Experimental Design

Teach the roles of:

independent variable;

dependent variable;

controlled variables;

confounders;

measurement uncertainty;

repeated trials.

Bad experiment:

```
change A
change B
change temperature
observe lamp
```

Better experiment:

```
hold B constant
hold temperature constant
change A only
measure lamp
```

Practical examinations should contain confounded experiments and test whether SEED recognizes that the result is inconclusive.

---

34. Data Analysis Tools

Provide safe mathematical instruments such as:

```
count
sum
mean
median
min
max
variance
frequency
histogram
correlation
```

The tools are not themselves understanding. SEED must learn which tool is useful for which research question.

Observational association and intervention-supported causal evidence must remain distinct.

---

35. Research Notebook UI

```
NURSERY / NOTEBOOK

Study #184
------------------------------------------------------------
QUESTION
Does Switch A control Lamp X?

OBSERVATIONS
#441 A off, B off -> X off
```

```
#442 A on,  B off -> X on

HYPOTHESES
H1 A controls X      support 3  contradictions 0
H2 B controls X      support 0  contradictions 2

EXPERIMENT
Changed: A
Controlled: B, room state
Measured: Lamp X

PREDICTION
H1 -> ON
H2 -> OFF

RESULT
ON

LIMITATIONS
3 trials
Switch C not yet tested

NEXT TEST
Vary C while A/B remain controlled

[Open evidence] [Replay] [Explain to caregiver]
```

Part VII - B15

36. B15 - Reading-to-Learn & Independent Study

Objective

Move from following written instructions to acquiring testable knowledge from text.

Target loop:

```
read explanation
 -> extract claim
 -> connect grounded concepts
 -> create predictions
 -> design practical test
 -> perform experiment
 -> compare result
 -> revise knowledge
```

Books become compressed experience from other minds, not automatic truth.

---

37. Reading-to-Learn Pipeline

```
TEXT
 -> token/construction model
 -> grounded concepts
```

```
-> claim extraction
-> source-provenance record
-> testability analysis
-> hypotheses/predictions
-> practical examination
-> observed results
-> belief/claim status update
```

Maintain explicit separation among:

external claim;

experiential belief;

experimentally tested conclusion.

---

38. Practical Examination

Example lesson:

"A slider controls a continuous value by moving a handle along a constrained path."

Training:

```
red horizontal slider
```

Exam:

```
blue vertical slider
```

If SEED succeeds only on the training layout, it memorized the example. If it succeeds on the new object, it demonstrates transfer/generalization.

Measure:

success;

primitive actions required;

prediction error;

interventions required;

generalization to new appearance/layout;

caregiver assistance.

---

39. Library / Reading UI

```
NURSERY / READING DESK / LIBRARY

Document                Read     Claims     Tested
-----------------------------------------------------
My First Controls v1       100%     12         10
Shape Guide v1             72%      8          4
Toy Terminal Manual        45%      14         6

Selected claim:
"Scrolling changes visible content without moving the pointer."
```

```
Status:
TESTED

Supporting experiments:
#811 #819 #824

Contradictions:
0

[Open source] [Open experiments] [Practical exam]
```

Part VIII - B16

40. B16 - Scientific Communication & Advanced Grounded Language

Communication develops throughout B11-B15. B16 focuses on clear evidence-backed explanation.

Communication progression

Early:

```
Observed:
Lamp ON

Changed:
Switch A

Inference:
A may affect lamp
```

Intermediate:

I changed A while B stayed off. The lamp changed from off to on. This supports H1, but only three trials have been completed.

Advanced:

```
Question
Background
Hypotheses
Method
Variables
Results
Analysis
Limitations
Conclusion
Future test
```

Natural-language prose should be generated from structured evidence-backed records rather than replacing those records.

---

41. Scientific Communication Metrics

Evaluate communication separately from discovery:

Factual fidelity: Does the explanation match stored evidence?

Evidence coverage: Did it include the important observations/results?

Inference separation: Did it distinguish observation from interpretation?

Uncertainty: Did it communicate limitations appropriately?

Completeness: Did it answer the actual question?

Clarity: Can the intended audience understand it?

Conciseness: Did it avoid irrelevant detail?

Traceability: Can substantive claims link back to evidence?

Audience adaptation: Can the same finding be explained at different technical levels?

A concise "I do not have enough evidence" should outperform a fluent unsupported explanation.

---

42. Language Algorithm Evolution

Recommended progression:

Language v0

n-grams;

variable-order Markov model;

context tree / PPM-style prediction.

Language v1

cross-situational grounding;

Bayesian association;

joint-attention weighting;

construction-template induction.

Language v2

grounded embeddings;

contrastive concept-to-experience learning;

clustering for larger vocabularies.

Language v3

small GRU or Transformer when justified by data and evaluation.

A neural language model must not become a replacement for:

world model;

body model;

evidence/provenance store;

scientific reasoning records;

decision authority.

Part IX - UI Component Inventory

43. Nursery Home Dashboard

```
+-------------------------------------------------------------------+
| NURSERY                                                           |
| Development environment                                           |
+------------------+-------------------+------------------------+
| CURRENT LESSON   | BODY              | COMMUNICATION          |
| Shape sorter     | 5 primitive actions | 18 grounded tokens   |
| Episode 41       | 2 skill candidates  | 4 tentative concepts |
| Exam mode: No    |                   |                        |
+------------------+-------------------+------------------------+
| RESEARCH         | READING           | CAREGIVER              |
| 3 hypotheses     | 2 documents       | 7 teaching events      |
| 2 experiments    | 8 claims          | 1 correction           |
|                  | 3 tested          |                        |
+------------------+-------------------+------------------------+
| RECENT DEVELOPMENT                                            |
| #841 reliable cursor +x consequence                           |
| #844 tentative "button" grounding                             |
| #850 failed practical transfer: vertical slider               |
|                                                               |
| [Learning Room] [Replay] [Evaluation]                         |
+-------------------------------------------------------------------+
```

Do not create a global "development level" percentage.

---

44. Global Nursery Header

Every Nursery view should expose:

```
Environment
Lesson
Version
Seed
Episode
Training / Exam mode
Pause
Resume
Reset
Replay
```

This makes the current experimental context impossible to confuse with general organism state.

---

45. Component Inventory

World

canvas;

virtual cursor;

buttons;

switches;

sliders;

lights;

shapes;

containers;

door/key;

drawing paper;

alphabet/number cards;

text fields;

toy terminal;

toy file browser;

scientific instruments.

SEED state panel

current observation;

current goal;

selected primitive/skill;

predicted result;

observed result;

prediction error;

relevant body beliefs;

relevant grounded words/constructions.

Caregiver

joint-attention selector;

teaching utterance;

demonstration recorder;

correction/clarification;

question prompts;

lesson controls.

Research

question board;

hypothesis board;

variable table;

experiment designer;

instrument rack;

trial table;

analysis tools;

limitations panel.

Reading

document reader;

known/unknown concept annotations;

claim extraction;

source provenance;

testability indicator;

practical exam launcher.

Replay

timeline;

step backward/forward;

observation;

action;

prediction;

environment transition;

teacher event;

knowledge update;

links to corresponding Astra evidence.

Part X - Headless Operation and Data

46. Headless Nursery Mode

Every lesson must run without the UI.

Conceptual command:

```
seed nursery run \
  --lesson button_discovery_v1 \
```

```
--seed 42 \
--episodes 100
```

The same environment engine should power:

interactive UI;

unit tests;

CI;

batch experiments;

practical exams;

replay.

The browser must never become the source of truth.

---

47. Lesson Specification

Conceptual example:

```yaml
lesson_id: button_discovery
version: 1
seed: 42

world:
  button:
    position: [400, 200]
  light:
    state: off

observations:
  - visible_objects
  - cursor
  - light_state

actuators:
  - move_cursor
  - mouse_down
  - mouse_up

goal:
  predicate: light.state == on

budgets:
  actions: 100
  wall_seconds: 300

hidden_rules:
  button_press_toggles_light: true
```

SEED must never receive hidden_rules. The evaluator may use them for scoring.

---

48. Practical Exam Specification

Training may use:

```
button position = left
button color = red
```

Evaluation may use:

```
button position = right
button color = blue
```

Same hidden affordance, different superficial features.

Later exams may deliberately transfer across object families to measure abstraction.

---

49. Conceptual Data Model for SEED 0.3

Do not finalize schema before formal batch design, but plan conceptual records such as:

```
nursery_sessions
environment_transitions
body_action_models
affordance_models
skills
skill_attempts
teaching_events
token_associations
constructions
external_claims
research_questions
hypotheses
experiments
experiment_trials
research_notes
```

Prefer append-only evidence plus versioned current projections.

Do not reinterpret or overwrite original B1 autobiographical memories to fit later concepts.

Part XI - Safety and Governance

50. Virtual-First Boundary

B11-B16 should initially operate only through:

deterministic virtual world;

toy terminal;

toy filesystem;

simulation APIs;

explicit caregiver interactions.

Not allowed during these early batches:

arbitrary host mouse/keyboard control;

real shell execution;

arbitrary host file access;

web browsing;

package installation;

privilege changes;

autonomous source modification;

self-installation/deployment.

---

51. Future Real-Body Bridge

Only after measurable virtual competence should a future separately authorized milestone consider:

```
NurseryBody
  -> learned skill representation
  -> PermissionedMacBody adapter
```

Each real actuator should require:

owner permission;

narrow action contract;

stop precedence;

audit;

rate limits;

dry-run or simulation where possible;

environment restrictions;

explicit separate authorization.

A later experiment may test whether a skill learned in simulation transfers to a real desktop. This is not part of B11.

Part XII - Evaluation Across SEED 0.3

52. Reuse B10's Experiment Engine

B11-B16 should not invent separate evaluation frameworks. Reuse the B10 scenario/runner protocol.

Body

actuator-effect prediction error;

evidence required for stable body model;

adaptation to new actuator behavior.

Affordances

interaction attempts;

success/failure rate;

transfer to unseen object.

Skills

primitive actions per task;

skill success;

skill reuse;

transfer/generalization.

Communication

referent accuracy;

instruction execution;

clarification rate;

grounding calibration.

Research

hypotheses proposed;

hypotheses eliminated per experiment;

confounder detection;

trials required;

adaptation after hidden-rule change.

Reading

instruction comprehension;

practical exam transfer;

claim verification rate;

unsupported-claim rejection;

source provenance completeness.

Scientific communication

factual fidelity;

evidence traceability;

uncertainty accuracy;

clarity;

audience adaptation.

Never collapse these into one universal intelligence score.

Part XIII - Batch Acceptance Milestones

53. B11 Milestone

Give SEED an unfamiliar virtual actuator.

Pass only if SEED can learn a reproducible action-to-consequence relationship from experience without an answer label.

54. B12 Milestone

Give SEED an unfamiliar object implementing a known affordance.

Pass only if a learned skill transfers beyond the original training object/layout.

55. B13 Milestone

Teach separately:

```
press
blue
button
```

Exam:

```
"Press the blue button."
```

Use an unseen layout.

Pass only if SEED composes grounded meanings and acts correctly without hidden answer injection.

56. B14 Milestone

Give SEED an unknown switch/lamp system.

Pass only if SEED:

distinguishes observations from inference;

forms competing hypotheses;

selects controlled experiments;

records predictions before tests;

revises hypotheses according to outcomes;

reports limitations.

57. B15 Milestone

Give SEED an unfamiliar simple reading passage such as:

"A toggle alternates between two persistent states when activated."

Then present an unseen toggle.

Pass only if SEED:

extracts the claim;

grounds or identifies unknown terms;

predicts behavior;

runs a practical test;

generalizes beyond the training example;

updates the claim's epistemic status from evidence.

58. B16 Milestone

Ask:

"What did you discover and why do you believe it?"

Pass only if substantive claims are traceable to evidence and uncertainty/limitations are communicated correctly.

Part XIV - Recommended Development Order

59. Phase 0 - Complete SEED 0.2

B10

Follow the existing design-first governance:

```
B10 design
-> owner review
-> implementation plan
-> owner review
-> explicit implementation authorization
-> implementation
-> local verification
-> independent review
-> hosted CI
-> merge
-> exact post-merge verification
```

Then freeze/tag SEED 0.2 as the developmental evaluation baseline.

---

60. Phase 1 - SEED 0.3 Foundation

B11

Implement only:

deterministic Nursery environment engine;

virtual body;

Learning Room;

lessons;

replay;

headless execution;

B10-compatible evaluation.

Do not implement full language or scientific inquiry yet.

---

61. Phase 2 - Skills

B12

Implement:

body schema;

affordance learning;

reusable skill representation;

Skills UI;

transfer evaluation.

---

62. Phase 3 - Communication

B13

Implement:

caregiver evidence;

grounded tokens;

transparent next-token/context models;

construction learning;

Reading-to-Act;

Vocabulary and Caregiver views.

---

63. Phase 4 - Research

B14

Implement:

observation/inference/hypothesis distinctions;

competing hypotheses;

controlled experiment planning;

Research Lab;

Notebook;

safe analysis tools.

---

64. Phase 5 - Reading

B15

Implement:

external claim records;

source provenance;

Reading-to-Learn;

practical examination;

independent-study workflow.

---

65. Phase 6 - Communication Depth

B16

Implement:

evidence-backed explanations;

research reports;

audience adaptation;

advanced grounded language modeling only when justified by data/evaluation.

Part XV - Explicit Non-Goals

66. What This Roadmap Must Not Become

Do not:

put a pretrained chatbot into Nursery and call that developmental language learning;

let a language model directly rewrite SEED's world model or evidence store;

expose the real shell as an early learning toy;

hard-code high-level skills as primitives and then claim SEED learned them;

let the browser DOM become the environment source of truth;

reward memorization of one training layout and call it generalization;

treat reading as truth without source/evidence status;

create one universal intelligence/development score;

start B11 while B10's evaluation protocol is unstable;

merge B11-B16 into one giant implementation batch.

Part XVI - Governance and Authorization

67. Batch Governance

This roadmap is owner approved as the current direction of the project.

It does **not** authorize implementation of B10 or any SEED 0.3 batch.

Every major batch continues to follow:

```
design / specification
-> owner review
-> implementation plan
-> owner review
-> explicit implementation authorization
-> implementation
-> independent review
-> verification
-> hosted CI / merge gates where applicable
```

The next authorized action after repository documentation is updated should be:

B10 DESIGN ONLY

B10 design must incorporate the reusable evaluation protocol in this roadmap while explicitly excluding Nursery implementation.

---

68. Final Strategic Statement

SEED 0.2 asks:

Can a bounded developmental organism learn from experience, and can that learning be measured honestly?

SEED 0.3 asks:

Can that organism learn how to use a body, discover what objects afford, compose skills, communicate with a caregiver, investigate an unknown world scientifically, learn from reading plus practical experience, and explain what it discovered with evidence and uncertainty?

The project should not aim to create a chatbot that happens to use tools.

The target is a developmental system whose language, skills, research behavior, and communication grow from interaction, prediction, evidence, experimentation, teaching, and verification.

---

69. Approved Next Step

Update canonical repository documentation to record B9 source/CI closure.

Add this roadmap to the repository as the owner-approved future-development direction.

Revise the canonical B10 description to **Development Evaluation & Experimental Foundation**.

Add the post-B10 SEED 0.3 roadmap as planned/not-started work.

Stop after documentation update and owner review.

After owner approval of the repository documentation update, begin **B10 design only** under a separate instruction.